

Clinical Trial Efficacy Evaluation

Statistical comparison of treatment and placebo groups using endpoint reduction

Dataset	Synthetic clinical trial dataset (educational use)
Participants	600 total; 300 Treatment; 300 Placebo
Primary endpoint	Reduction = BaselineScore – Week12Score
Primary test	Welch independent-samples t-test
Non-parametric check	Mann–Whitney U test
Significance level	$\alpha = 0.05$

1. Objective

To evaluate whether the treatment group showed a statistically different reduction in the primary clinical endpoint at Week 12 compared with the placebo group.

2. Methodology

For each participant, endpoint reduction was calculated as BaselineScore minus Week12Score; therefore, a larger positive value represents a larger reduction. The treatment and placebo groups were compared using Welch's independent-samples t-test, which does not require equal group variances. A two-sided Mann–Whitney U test was also performed as a non-parametric sensitivity check. Statistical significance was assessed at $\alpha = 0.05$.

3. Descriptive Results

Study arm	N	Mean reduction	SD	Median
Placebo	300	3.42	6.85	3.15
Treatment	300	9.23	7.56	9.35

The treatment group had a mean reduction of 9.23 points versus 3.42 points in the placebo group. The estimated mean difference (Treatment – Placebo) was 5.81 points, with a 95% confidence interval of 4.66 to 6.97.

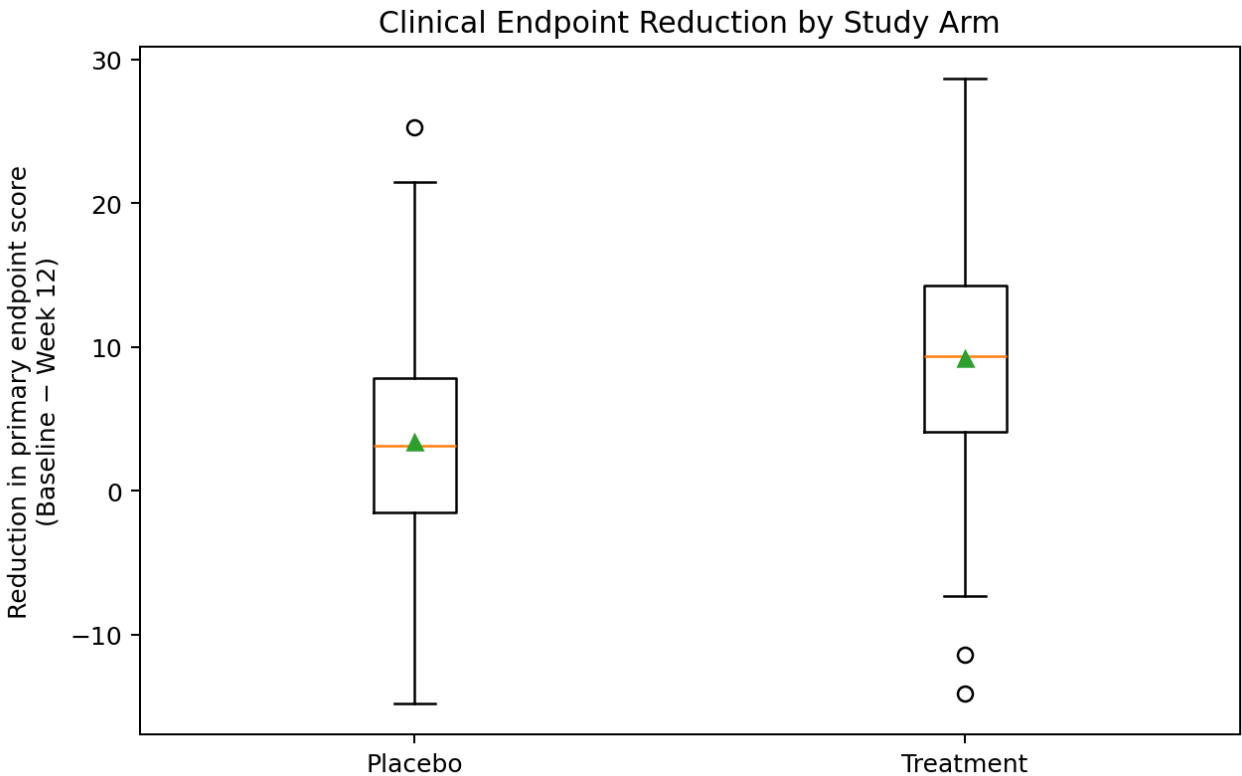


Figure 1. Distribution of endpoint reduction in the two study arms.

4. Statistical Testing

Test	Statistic	df	p-value	Interpretation
Welch independent t-test	9.870	592.2	2.255e-21	Statistically significant
Mann–Whitney U	64776.5	—	1.222e-20	Statistically significant

5. Confidence Interval

The 95% confidence interval for the treatment-minus-placebo mean difference was 4.66 to 6.97 points. Because the interval is entirely above zero, the estimated treatment effect is positive for this synthetic dataset.

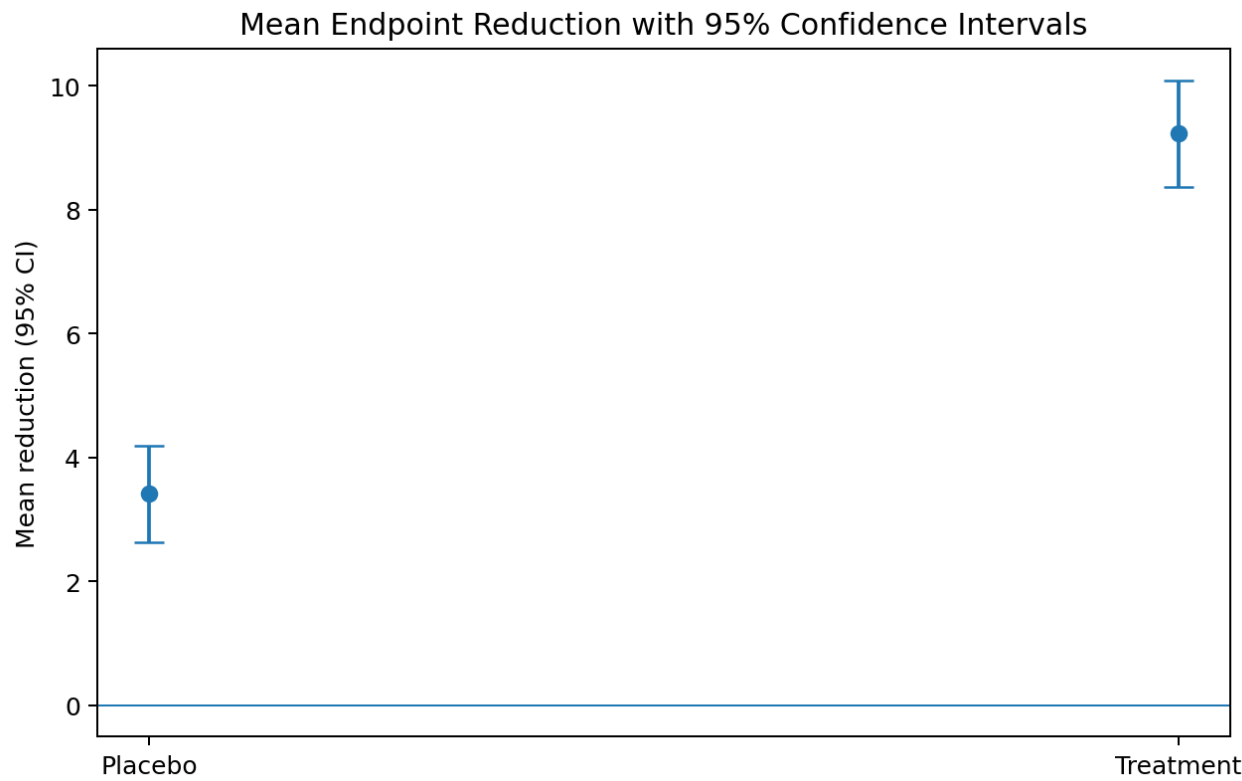


Figure 2. Mean endpoint reduction with approximate 95% confidence intervals for each group.

6. Clinical Efficacy Summary

In this educational synthetic dataset, participants receiving treatment showed a greater mean reduction in the primary endpoint than participants receiving placebo (9.23 vs 3.42 points). The Welch t-test produced $t = 9.87$, $df = 592.2$, $p < 0.001$, indicating a statistically significant between-group difference at the 5% level. The Mann–Whitney U analysis provided a consistent result ($U = 64776.5$, $p < 0.001$). These findings support evidence of a treatment-associated endpoint reduction within this synthetic educational dataset; they should not be interpreted as evidence of real-world clinical efficacy because the dataset is synthetic and intended for statistical learning.

7. Limitations

- The dataset is synthetic and intended for educational statistical analysis.
- The analysis compares endpoint change scores and does not adjust for baseline covariates.
- Statistical significance does not by itself establish clinical importance or benefit-risk balance.
- No causal or regulatory conclusion should be drawn from this exercise.

8. Reproducibility

The accompanying statistical-results CSV contains the key calculated values. The analysis uses the uploaded `clinical_trial_efficacy_v2.csv` dataset and the definitions stated above.